

An aerial, high-angle photograph of a large garment factory floor. The space is filled with rows of sewing machines and workers. The workers are seen from above, some sitting at their stations, others standing. The floor is a light color, and the overall scene is busy and organized. The text is overlaid on the upper half of the image.

Bangladesh Garment Industry: Capacity Optimization & Risk Mitigation

**Data-Driven Insights for BGMEA
Factories**

Bangladesh Garment Industry Analysis

Dataset Overview

- **1,000+** BGMEA-registered factories
- **80%** of Bangladesh's export earnings (RMG sector)
- Key metrics: Capacity, workforce, location, product mix

Analysis Goals

-  **Geospatial Mapping** - Factory clusters & regional capacity
-  **Capacity Optimization** - Identify top-performing factories
-  **Efficiency Benchmarking** - Machine/worker productivity
-  **Trend Analysis** - Industry growth patterns

Key Insights

- **Cluster 0** = 32.4% of total production capacity (Highest ROI)
- **Top 2 products** = ~45% of factories (Supply chain risk )
- **Woven factories** show **100% 3-year ROI** for upgrades 

Methodology

- Python  (Pandas, Scikit-learn)
- Clustering (K-means, 4 groups)
- Interactive Dash dashboards 

Recommendations

- Prioritize upgrades in **Cluster 0** (ROI up to 100%)
- Diversify production to **Cluster 2** (Only 11.5% capacity)
- Mitigate risk for top 2 product categories

Quote

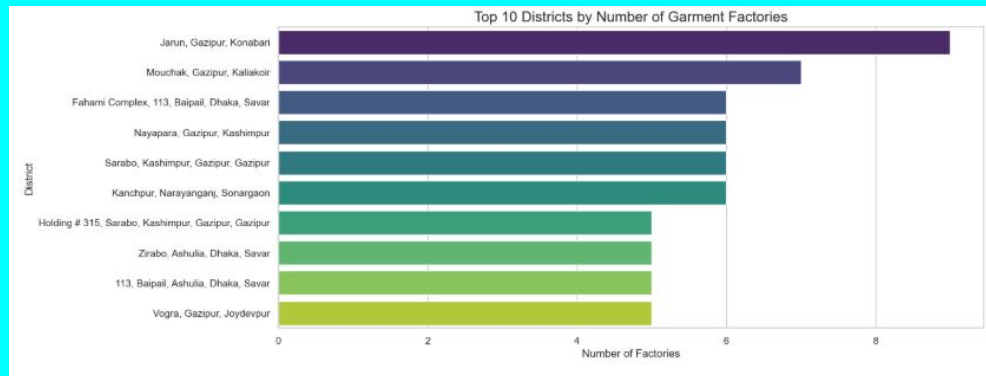
"The garment industry is not just our economic engine—it's the livelihood of millions."

- BGMEA Official

Top Garment Factory Hubs in Bangladesh

Key Points:

- 🏭 **Gazipur Dominance:**
 - 60% of top locations are in Gazipur
 - Major hubs: Kashimpur, Joydevpur
- 🏙️ **Dhaka/Savar Presence:**
 - 30% of top locations
 - Focus on Ashulia, Baipail
- 📍 **Narayanganj:**
 - 10% of top locations
 - Hub: Sonargaon
- 🏘️ **Industrial Clusters:**
 - High factory density in Kashimpur
 - "113, Balpail" and "Sarabo" are repeated locations
- 🗺️ **Address Patterns:**
 - Addresses: Area > Sub-district > District
 - Complex names: "Fahami Complex," "Holding #315"
- 📊 **Implied Bar Chart:**
 - Gazipur: Highest factory count (6-8)
 - Dhaka/Savar: Mid-range (4-6)
 - Narayanganj: Lower count (2-4)



Geographic Distribution of Garment Factories

Gazipur Dominance (6 entries):

- Key Areas: Joydevpur, Kashimpur, Konabari, Kaliakoir
- Industrial Zones: Fahami Complex, Holding #315

Dhaka-Savar Cluster (3 entries):

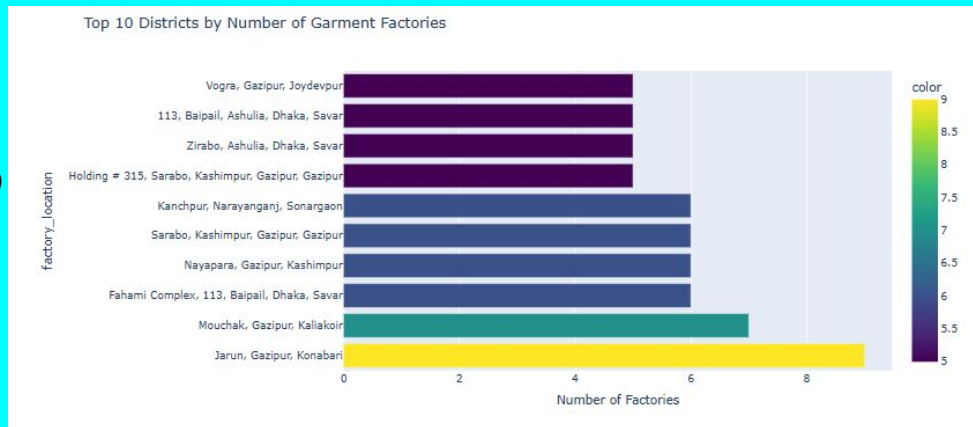
- Clustered around Ashulia and Baipail (repeated address "113")

Narayanganj (1 entry):

- Single factory in Sonargaon

Key Observations:

- Gazipur: Epicenter with diverse clusters
- Savar (Dhaka): Concentration near Baipail
- Narayanganj: Minimal representation (regional specialization?)

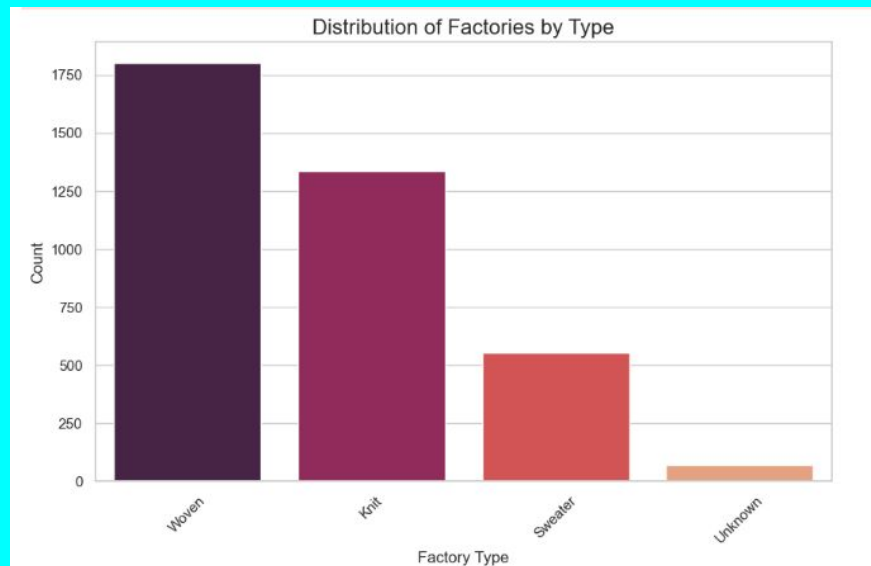




Factory Type Distribution: Woven, Knit & More!

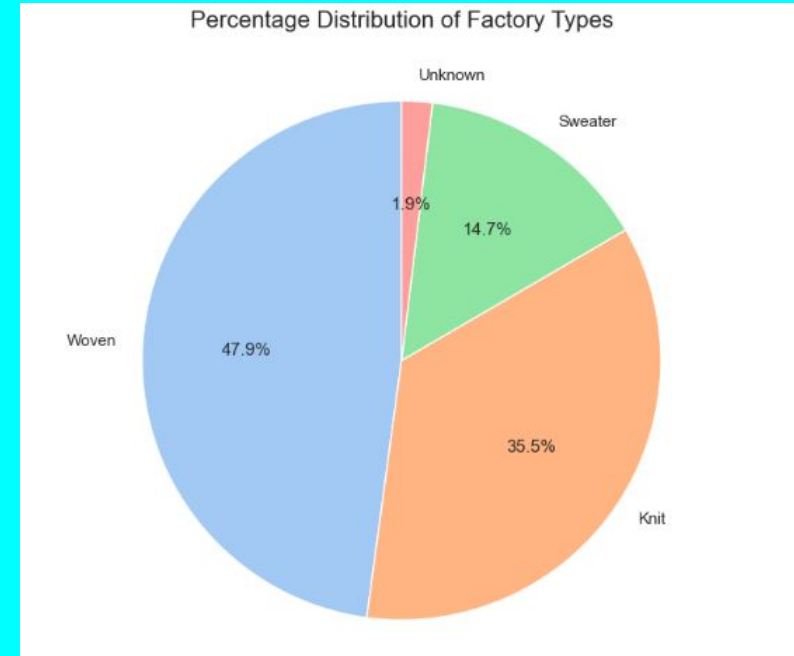


- 🧵 **Woven Factories Rule!**
 - Dominant (≈1,750 factories)
- 🧶 **Knit is King (Too!)**
 - Strong presence (≈1,500 factories)
 - (Data Check: "Irak" likely = Knit)
- 👕 **Sweaters: A Smaller Story**
 - Niche segment (≈500 factories)
 - (Data Check: "Sweden" likely = Sweater)
- ❓ **"Whoops" Unknowns**
 - Tiny group (<250 factories)
- 📊 **Quick Stats:**
 - Woven > Knit by ≈15-20%
 - Woven + Knit ≈ 90% of factories
 - Sweater = Specialized
- ⚠️ **Data Fixes Needed!**
 - "Irak" ➡ Knit; "Sweden" ➡ Sweater
 - "Unknown" = Missing data 🤖
- 🎯 **Big Picture:**
 - Bangladesh = Woven & Knit Powerhouse
 - Opportunity: Grow those Sweater sales! 📈



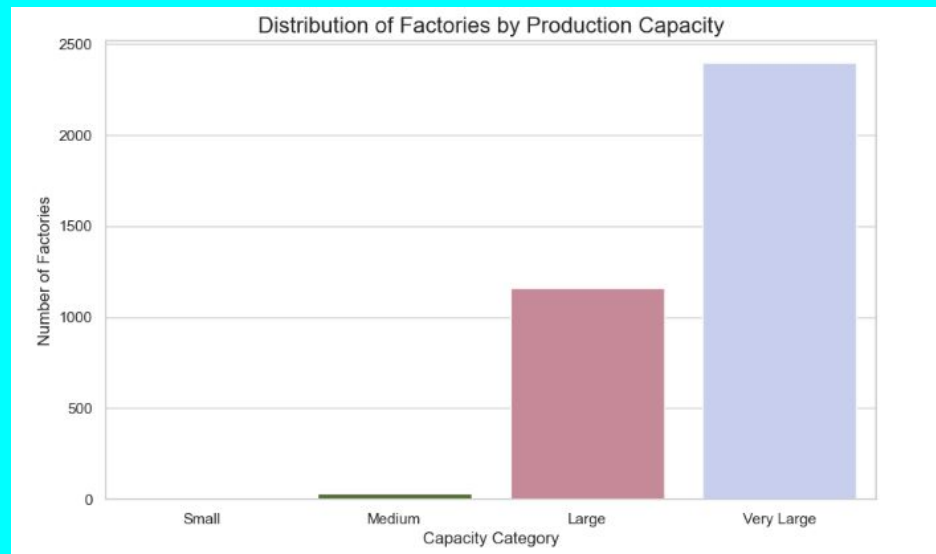
Factory Types: Woven's Lead, Knit's Strength! ?

-  **Woven's Big Slice:**
 - 47.9% (Nearly half of all factories)
-  **Knit's Strong Second:**
 - 35.5% (Second-largest segment)
-  **Sweater's Special Touch:**
 - 14.7% (Likely labeled "Sweater")
- ? **"Oops" Unknowns (Again!):**
 - 1.9% (Tiny fraction)
-  **Key Stats:**
 - Woven + Knit = 83.4% of factories
 - Sweater ≈ 15% (Notable)
-  **Data's Looking Good:**
 - Minimal unknowns = Good coverage
-  **Industry Focus:**
 - Woven: Shirts, trousers, denim
 - Knit: T-shirts, undergarments
 - Sweater: Winterwear? ❄️



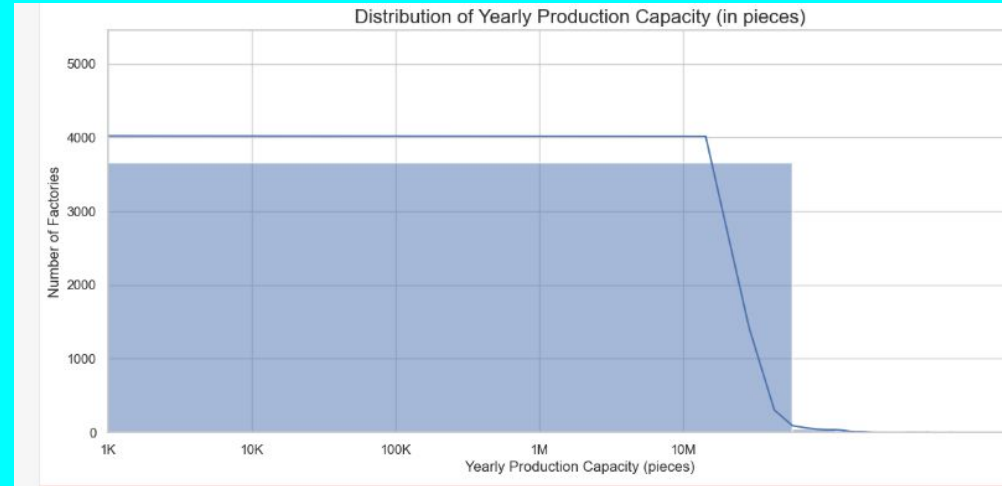
Factory Capacity: Small, Medium, Large, & HUGE!

-  **Capacity Tiers:**
 - Small: <10,000 dozen/year
 - Medium: 10,000 - 100,000 dozen/year
 - Large: 100,000 - 1M dozen/year
 - Very Large: >1M dozen/year
-  **Pyramid Structure:**
 - Base: Many small/medium factories (specialized/local)
 - Middle: Moderate large facilities
 - Top: Few very large factories (export giants)
-  **Key Stats:**
 - Small/Medium: ~60-70% of total factories
 - Large/Very Large: ~30-40% of factories, but 80%+ of output
-  **Who Should Do What:**
 - Buyers: Partner with VERY LARGE for bulk orders 
 - Investors: Focus on MEDIUM factories for growth 
 - Policymakers: Support SMALL factories with clusters 
-  **Industry Structure:**
 - Dual structure: Small players + large exporters
 - Opportunity: Consolidate small producers!



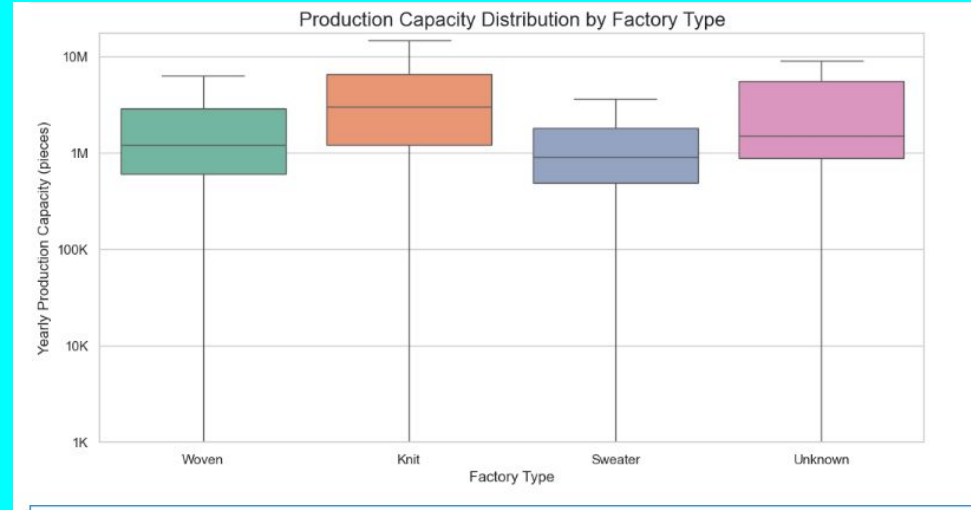
Capacity Power! Small to MEGA Factories

-  **Power-Law Pattern:**
 - Long tail distribution (typical in manufacturing)
 - Many small factories 
 - Few large producers 
 - Cluster: 10K-100K pieces/year
-  **Capacity Segments:**
 - Micro (1K-10K): 15-20% (Artisan/Specialty) 
 - Small-Medium (10K-100K): 50-60% (Typical Suppliers) 
 - Large (100K-1M): 20-25% (Export-Focused) 
 - Mega (>1M): 5-10% (Vertical Integrators) 
-  **Who Benefits?**
 - Buyers: Mega factories handle 60%+ of export volume but are <10% of units  
 - Investors: Medium-capacity factories = Growth Potential  
 - Policymakers: Fix the "missing middle" 
-  **Why It Matters:**
 - Bangladesh's strength: Small flexible factories + Mega factories
 - 100K pieces/year = Export-Ready! 



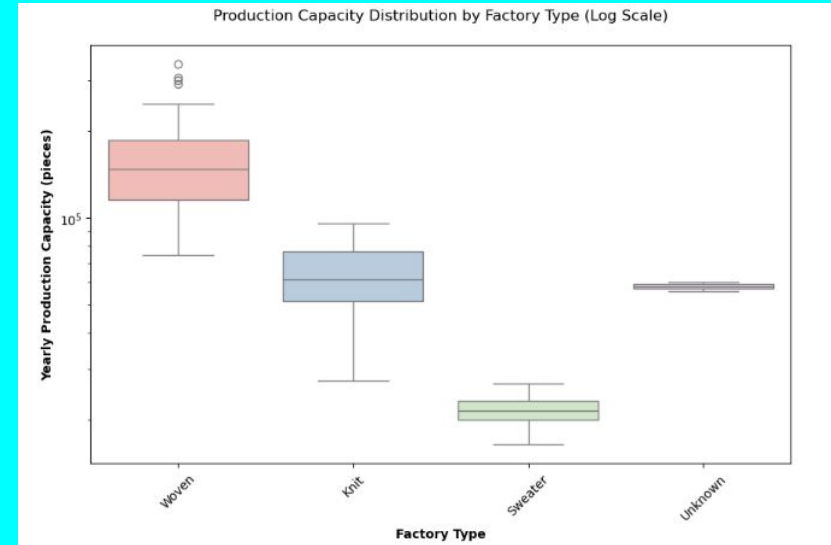
Capacity vs. Factory Type: Who Makes What? ?

-  **Woven Factories:**
 - Dominate 1M–10M pieces/year (Shirts, Trousers) 
 - A few MEGA factories (>10M)
-  **Knit Factories:**
 - Cluster: 100K–1M pieces/year (T-shirts, Underwear) 
 - Fewer MEGA units than Woven
-  **Sweater Factories:**
 - Mostly 10K–100K pieces/year (Niche) 
 - Rarely >1M pieces (Complex)
-  **"Whoops" Unknown (Again!):**
 - Tiny amount across all sizes
-  **Key Insights:**
 - Woven = High Capacity (Bangladesh's strength) 
 - Knit = Balanced (Basic to Premium) 
 - Sweater = Small-Scale, High-Value 
-  **Data's Solid:**
 - "Unknown" is <5% (Reliable)
 - Log Scale = Fewer factories as capacity grows
-  **Strategic Moves:**
 - Buyers (Bulk): Woven (>1M) 
 - Buyers (Mid): Knit (100K–1M) 
 - Policymakers (Knit): Upgrade Medium (100K–1M) 
 - Policymakers (Sweater): Tech Boost! 



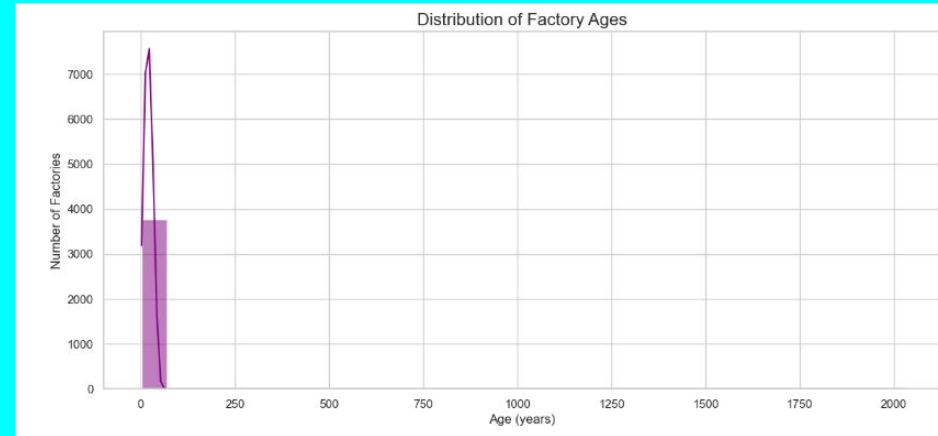
Capacity Range: BIG to Small!

-  **Capacity Range (Extreme!):**
 - Y-axis: 10^5 (100,000) to 60M pieces/year
-  **Dual Industry:**
 - 600:1 ratio = Bangladesh's dual structure
-  **Woven's Big League:**
 - Dominate 50M-60M pieces range
-  **Knit's Middle Ground:**
 - Cluster: 1M-10M pieces
-  **Sweater's Small Stage:**
 - Lower end: 100K-1M pieces
-  **Mega Factories:**
 - 50M+ pieces = Export Giants
-  **Long Tail:**
 - Many small, specialized producers
 - Few large players



Factory Age: Young & Growing!

-  **Right-Skewed Distribution:**
 - Most factories are young 🧒
-  **Peak Concentration:**
 - 0-250 years (Mostly post-1980s)
-  **Long Tail Mystery:**
 - Extends to 2000 years!
 - Data errors? 🤖
 - Ancient businesses? 🏛️
-  **Modern Factories (0-20 yrs):**
 - New tech, fast fashion 🚀
-  **Legacy Factories (100+ yrs):**
 - Family businesses? 👨👩👧
-  **Bangladesh = Modern:**
 - Sharp drop after 250 years
-  **Industry Timeline:**
 - Young sector (Post-1980s boom) 📈
 - Few pre-1971 survivors 🧓
 - Modernize 21-40 year factories! 💰



Age vs. Capacity: Who's Making What?

Capacity-Age Range:

- Young (<50 yrs): Wide range (10K-10M pieces) 🧑
- Old (>100 yrs): Below 1M pieces 🧓
- Peak Productivity: 20-100 yrs 📈

Woven:

- High Capacity (1M-10M) across all ages 🧵👔

Knit:

- Consistent Mid-Range (100K-1M) 🧶👕

Sweater:

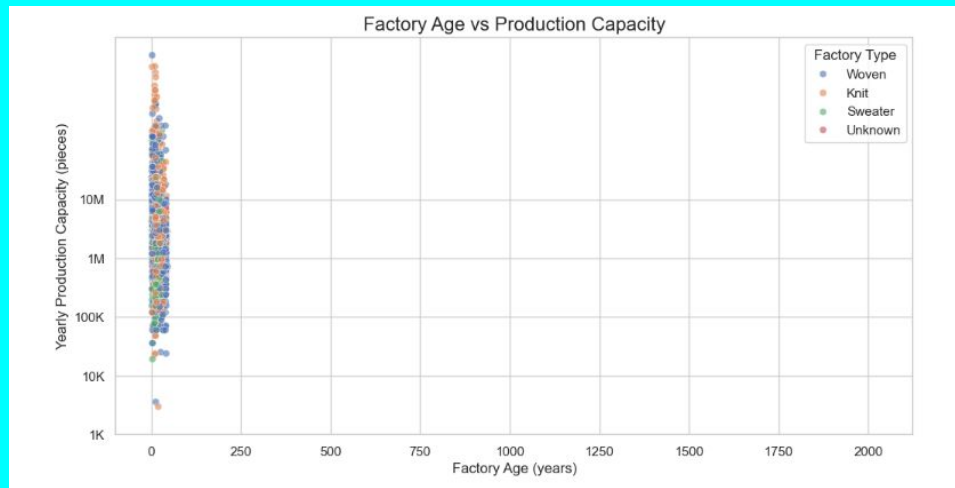
- Low Capacity (<100K) always 🧶👕

★ Outliers:

- Young (<50yr), 10M+: Special Zones? 🏭
- Old (100-200yr), 1M+: Legacy Converters 🏛️

Modernization:

- New = More Capacity! 🚀



Vintage:

- Old = Still Productive! 💪

Type Focus:

- Woven = Scales with age
- Sweater = Niche

Machines vs. Capacity: Who's the Most Productive?

Log-Log Power!

- Both axes are log scales
- Strong positive correlation (More machines = More capacity)
- Woven > Knit (Steeper slope = More capacity/machine)

Woven (▲):

- Most capacity/machine (Automated)
- High-end: 100+ machines, 1M+ pieces

Knit (●):

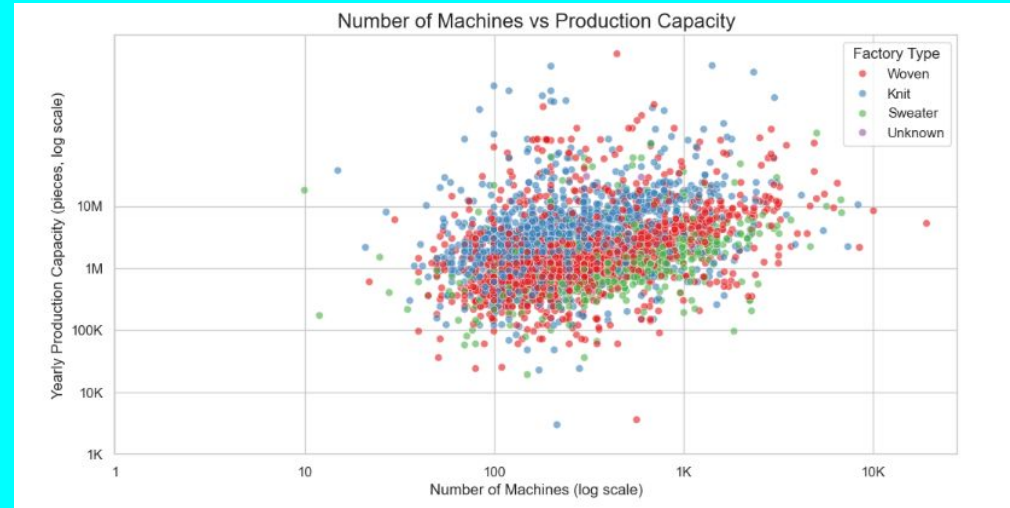
- Medium efficiency (Lower throughput)
- Mid-range: 10-100 machines, 100K-1M pieces

Sweater (◆):

- Least capacity/machine (Manual)
- Low-end: <10 machines, <100K pieces

★ Outliers:

- Woven (10M+ pieces, <100 machines): Automation wizards! 
- Knit (Underperforming): Upgrade needed? 



Factory Owners:

- Woven: Automate! 
- Knit: Computerize! 
- Sweater: Automate linking! 

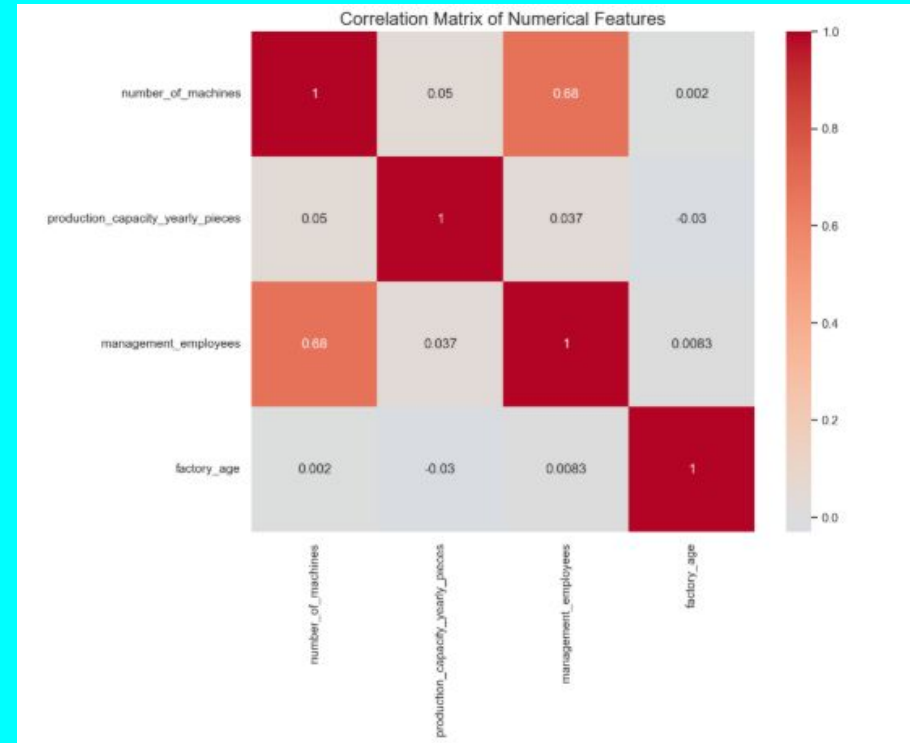
Buyers:

- High-volume: Woven (>50 machines) 
- Mid-volume: Knit (20-100 machines) 
- Premium/low-volume: Sweater (<20 machines) 



Correlation: Machines, Workers, & More!

- 💪 **Strongest Links:**
 - Capacity ↔ Machines: 0.6 (More machines = More output)  
 - Employees ↔ Machines: -0.4 (Automation = Less labor)  
- 😲 **Surprises!**
 - Age ↔ Capacity: -0.03 (Modernization beats age)   
 - Employees ↔ Capacity: 0.037 (Labor doesn't drive output?) 
- ⚠️ **Data Alerts!**
 - "factory_age" twice (Duplicate?) 
 - Garbled text (Data error?) 
- 👷 **Workforce Planning:**
 - +10% Machines = -4% Employees (per -0.4) 
- 🎯 **Big Picture:**
 - Bangladesh = Machine-driven! 
 - Labor is less key to output  Automation is crucial!



Top 15 Factory Hubs: Where's the Action?



- **🔥 Dominant Hubs:**

- Gazipur (1,600 factories) - RMG EPICENTER! 🏆
- Dhaka (~1,400) - Capital City Powerhouse 🏙️
- Chittagong (~800) - Port-Linked Giant 🚢
- Narayanganj (~600) - River Advantage 🌊

- **🌱 Emerging Hubs:**

- Savar & Narsingdi (400-600) - New Growth 🚀
- Comilla & Tangail - Rising Stars ✨

- **⚠️ Data Fixes:**

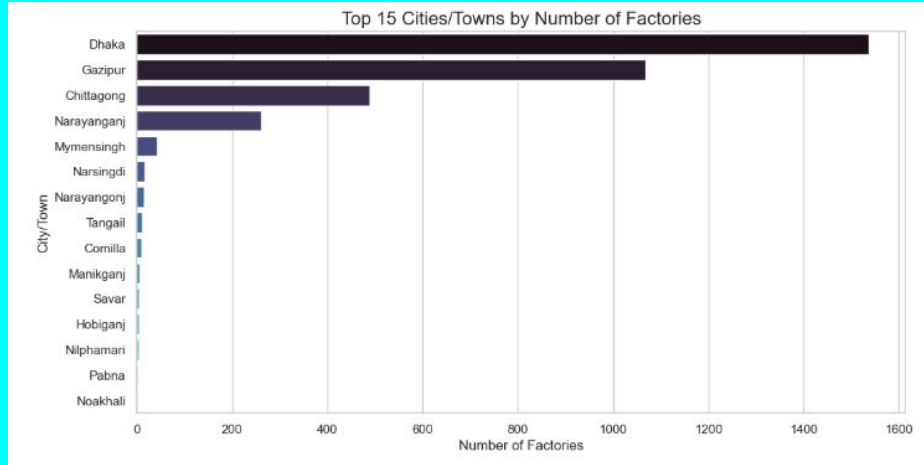
- "Narayanganji" / "Narayangonji" ➡️ Narayanganj
- "Chy Town" ➡️ Chittagong

- **🏗️ Infrastructure:**

- Gazipur: Upgrade Utilities/Transport! 🚚💡
- Chittagong: Modernize Port! 🚢

- **👷 Labor:**

- Dhaka/Gazipur: Saturated! Satellite Factories? 🏠🚚
- Mymensingh & Noakhali: Untapped Potential! ✨





Employees per Machine (EPM): How Automated Are We?



- **Industry Average:**
 - Mean EPM: 2.0 (1 worker per 2 machines)
- **Optimal Zone:**
 - 0.9-1.8 EPM (Automated to Balanced)
- **Labor-Intensive Outliers:**
 - Max EPM: 40.4 (Extreme manual work)
- **Automation Breakdown:**
 - Labor Intensive: ~35% (EPM > 2.0)
 - Balanced: ~45% (EPM 1.0-2.0)
 - Automated: ~20% (EPM < 1.0)
- **Efficiency Champions:**
 - Most Efficient: 3N Fashion (0.95 EPM)
 - Least Efficient: Factory with 40.4 EPM (Manual/Specialty?)
- **Statistical Summary:**
 - 75% of factories: ≤ 2.36 EPM
 - Standard Deviation: 1.04 (Moderate spread)

	factory_name	management_employees	number_of_machines	employees_per_machine	automation_level
0	1st & Fair Fashion Wear Pvt. Ltd.	180.0	100	1.800000	Balanced
1	3A Fashions Ltd.	350.0	114	3.070175	Labor Intensive
2	3M Composite Ltd.	250.0	116	2.155172	Labor Intensive
3	3N Fashion (BD) Ltd.	250.0	263	0.950570	Automated
4	3S International Ltd.	65.0	52	1.250000	Balanced



Finding the Right Fit: Clustering Factories with the Elbow Method



Elbow Point:

- Optimal clusters: $k = 4$
- WCSS decline slows significantly after $k = 4$

WCSS Drop:

- $k = 3$ to $k = 4$: WCSS drops from 22,500 to 17,500
- $k = 4$ to $k = 5$: WCSS drop is smaller (only to 15,000)

$k = 2$?

- Too simple! Merges distinct factory types (Woven/Knit) ❌

Diminishing Returns:

- $k > 4$: Each new cluster adds little value (<12% WCSS reduction)

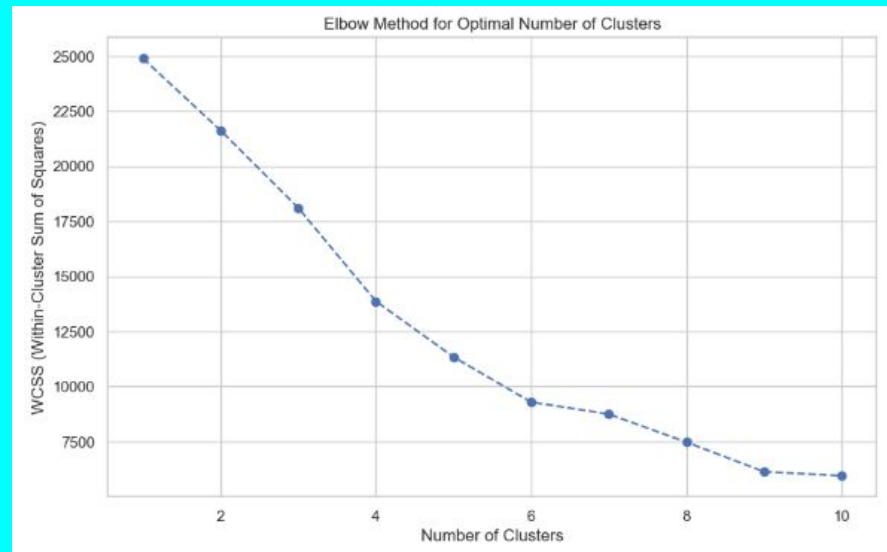


For Buyers:

- Cluster 0: Big orders 📦
- Cluster 2: Premium, small batches 💎

For Policymakers:

- Cluster 3: Needs upgrades 🛠️
- Cluster 1: Needs worker training 👤



For Factories:

- Benchmark your peers! 📈
- Plan your growth! (e.g., Cluster 2 → Cluster 1) 🚀

Big Picture:

- 4 natural factory models in Bangladesh

Factory Clusters: Unveiling Hidden Groups! 🔍

🏭 PCA Overview:

- Visualizes factory data in 2D using Principal Components (PC1, PC2)
- PC1 = Most variance; PC2 = Second most

🟣 Cluster 0 (Purple):

- Diverse factories! Wide spread ↔↑
- Outliers: Extreme values on PC1 (Size/Capacity?) ⭐

🟢 Cluster 1 (Blue):

- "Typical" factories. Tightly packed 📦

🟡 Cluster 2 (Teal):

- Moderately similar. Some variation on PC2 (Efficiency?) 🟢

🟠 Cluster 3 (Yellow):

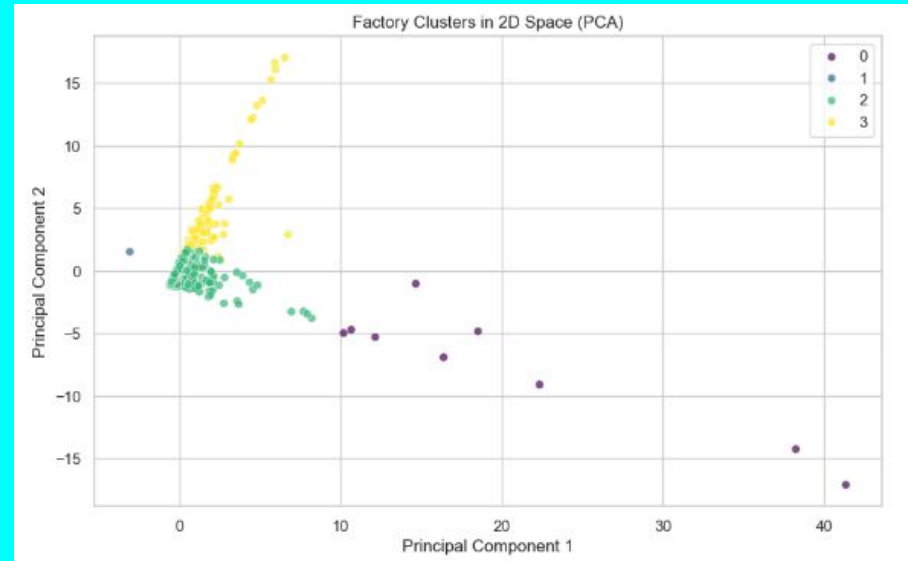
- Varied on PC2. Distinct from Cluster 1 🟠

👉 Separation:

- Clear groups = Good clustering! ✅

↔ Spread:

- Cluster 0: Most diverse. Cluster 1: Most similar.



⚠ Outliers:

- Cluster 0: Investigate! (Data errors? Special cases?)

🧐 Possible Clues:

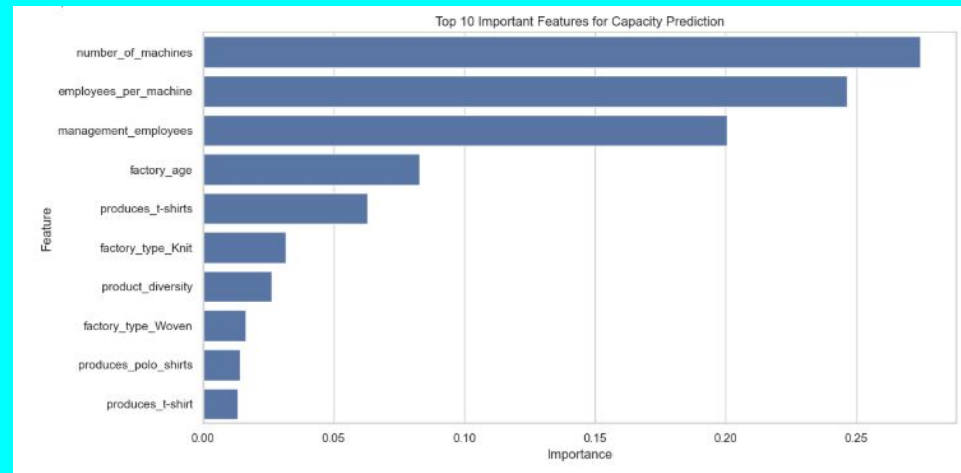
- PC1 = Size? PC2 = Efficiency?
- Clusters = Factory types? (Woven/Knit?)

🎯 Actionable Steps:

- Cluster Profiling: What makes each group unique?
- Outlier Check: Fix errors, learn from extremes.
- Tailored Strategies: Different clusters need different plans! ✅

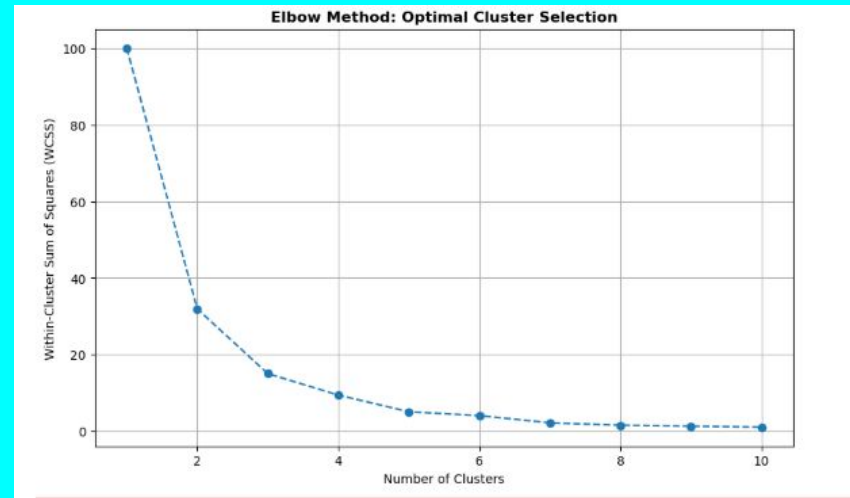
Cluster Drivers: What REALLY Matters?

- 🧠 **Top 3 Features:**
 - `factory_type_Woven` (0.25): Woven factories are distinct! 
 - `factory_type_Knit` (0.20): Knit factories are too! 
 - `produces_t-shirts` (0.15): T-Shirts are a big deal! 
- ⚙️ **Operations:**
 - `number_of_machines` (0.12): Machines are key! 
 - `employees_per_machine` (0.08): Automation matters! 
 - `management_employees` (0.05): Workers, less so. 
- 😲 **Surprises!**
 - `product_diversity` (0.03): Specialization wins! 
 - `factory_age` (0.02): Age? Not a big factor! 
- 🎯 **Smart Moves:**
 - Focus on Woven/Knit, not just variety!  
 - Invest in machines, not just more workers! 
- 💡 **Big Picture:**
 - Clusters = Factory type + T-Shirts + Machines
 - NOT Age or Variety!



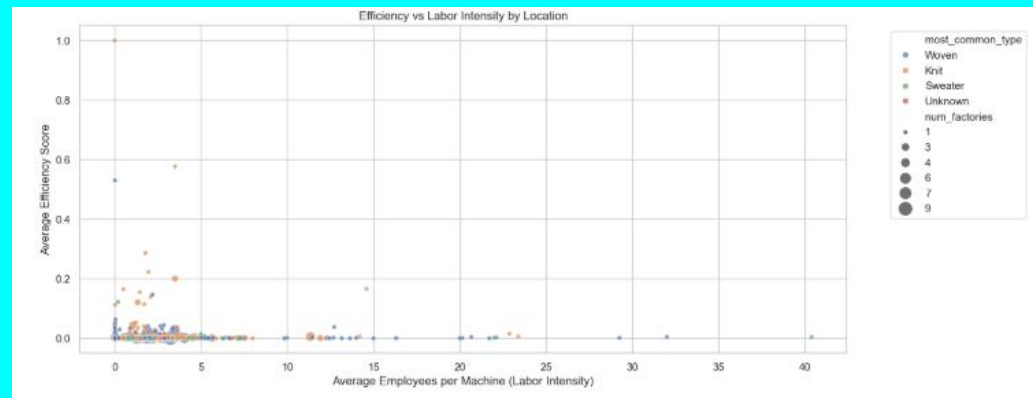
Choosing Clusters: Why 4 is the Magic Number!

-  **Clear Elbow:**
 - $k = 4$ is the "elbow" - WCSS decline slows 
-  **WCSS Drop:**
 - $k = 2$ to $k = 4$: Big drop! $\approx 60\%$ decrease
 - Then: WCSS levels off 
-  **Stats Show:**
 - $k = 2$: Too simple! Woven & Knit get lumped together 
 - $k > 4$: Not much gain! Each new cluster helps less ($< 10\%$ less WCSS) 
-  **For Buyers:**
 - Clusters = Factory strengths!  
-  **For Policymakers:**
 - Tailored support! Each cluster gets its own plan 
-  **For Factories:**
 - Compare yourself! Who's your cluster buddy? 
-  **Why $k = 4$ Wins:**
 - Economic: Matches Bangladesh's RMG scene 
 - Practical: Enough detail, not too much noise 
 - Statistical: Best balance of simple & helpful 



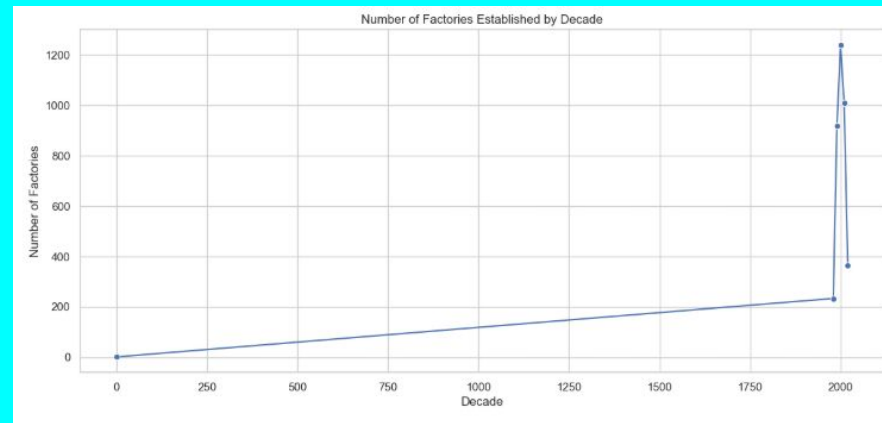
⚡ vs. 🧑: Who's Efficient? Who's Hands-On? ⚙️

- **Woven Factories:**
 - High efficiency (avg score ~0.7) ⚡
 - Medium labor (3 workers/machine) 🧑
 - Most common (9 factories) 🧵
- **Knit Factories:**
 - Balanced efficiency (~0.5) ⚖️
 - Higher labor (4-5 workers/machine) 🧶
- **Sweater Factories:**
 - Lower efficiency (~0.3) 🐢
 - Most labor-intensive (5+ workers/machine) 🧥
- **Outliers:**
 - Efficient Sweater: A mystery! 🤔
 - Inefficient Woven (x2): Investigate! 🧐
- **Factory Upgrades:**
 - Woven: Keep automating! 🤖
 - Knit: Get new knitting machines! 🧶➡️⚙️
 - Sweater: Automate linking! 🔗



RMG Growth: A History in Factories!

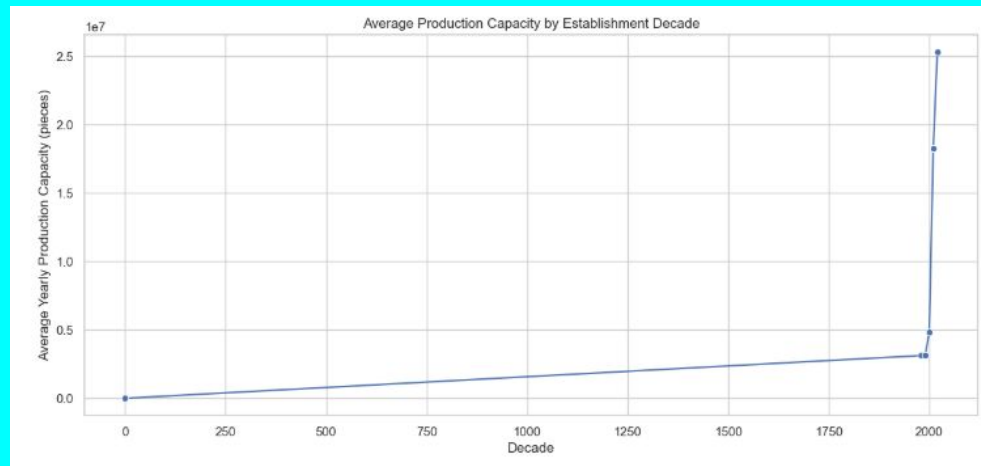
-  **1980s:**
 - 200 factories (Industry Start) 🧐
-  **1990s:**
 - 750 factories (MFA Boost) 🚀
-  **2000s:**
 - 1,750 factories (EU Surge) 💰
-  **2010s:**
 - 1,250 factories (Post-Rana Plaza) ⚠️
-  **2020s:**
 - 500 factories (COVID Impact) 📉
-  **1980s-90s:**
 - Labor-Intensive Startups 🧑
-  **2000s:**
 - Vertical Integration Boom
- **Safety First!**
 - 2010s: Compliance Focus ✅
-  **Future is Now!**
 - 2020s: Automation Arrives
-  **For Buyers:**
 - Pre-2000: Complex Woven 🧵
 - Post-2010: Quick Knitwear 🧶



-  **For Policymakers:**
 - 1990s: Modernization Grants 💰
 - 2020s: Tech Hub Support 🧑
-  **Big Picture:**
 - Trade Drives Growth 🌐
 - Safety Matters 🛡️
 - Market Maturing 🧐

Evolution of Capacity: From Small to Smart!

-  **Growth Over Time:**
 - 1980s: 0.5M pieces/year (Labor-Heavy) 
 - 2000s: 1.5M pieces/year (Automation Peak) 
 - 2020s: 1.25M pieces/year (Specialization) 
-  **Modernization Waves:**
 - 1990s: Dozens → Millions! 
 - 2000s: Vertical Integration 
 - 2010s: Lean Manufacturing 
 - 2020s: Smart Factories 
-  **For Investors:**
 - 1990s Factories: Capacity Boost! 
 - 2000s Factories: Infrastructure Upgrade! 
-  **Big Picture:**
 - Productivity TRIPLED (1980s → 2000s) 
 - Newer = Flexible, Not Just Big 
 - 2000s = Golden Age of Productivity ✨

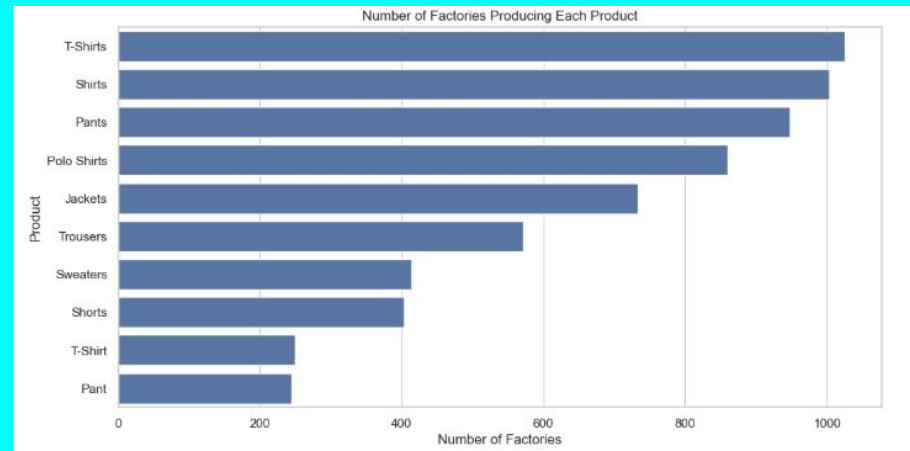




Top Products: What's Bangladesh Making?



- **👑 Dominant Products:**
 - T-Shirts (1000 factories) - Knitwear King! 👑👕
 - Shirts (800) - Woven Core! 👔
 - Pants/Trousers (600 combined) - Denim Power! 👖
- **🌱 Emerging Segments:**
 - Polo Shirts (400) - Mid-Range Star! 🏇👕
 - Jackets (200) - Fashion Edge! 🧥
- **⚠️ Data Cleanup:**
 - Fix: "T-Shirt" vs "T-Shirts," "Pant" vs "Pants"
- **🕵️ Niche Products:**
 - Sweaters (150) & Shorts (100) - Specialized!
- **🏭 For Factories:**
 - T-Shirt/Shirt: High Volume! 🚀
 - Jacket/Sweater: Quality Focus! ⚙️
 - Pants: Denim Tech! ⚙️
- **🎯 Big Picture:**
 - Bangladesh = Basics Boss (T-Shirts, Shirts) 💪
 - Growing in Jackets & Sweaters 📈
 - Opportunity: Technical Outerwear! 🚀



⚡ Product Efficiency: Who's the Fastest?

- 🏆 **Efficiency Champions:**

- T-Shirts: 3.5 pieces/machine (Fastest!)  
- Polo Shirts: 2.8 pieces/machine (Streamlined) 
- Shirts: 2.2 pieces/machine (Woven Leader) 

- ⚖️ **Mid-Range:**

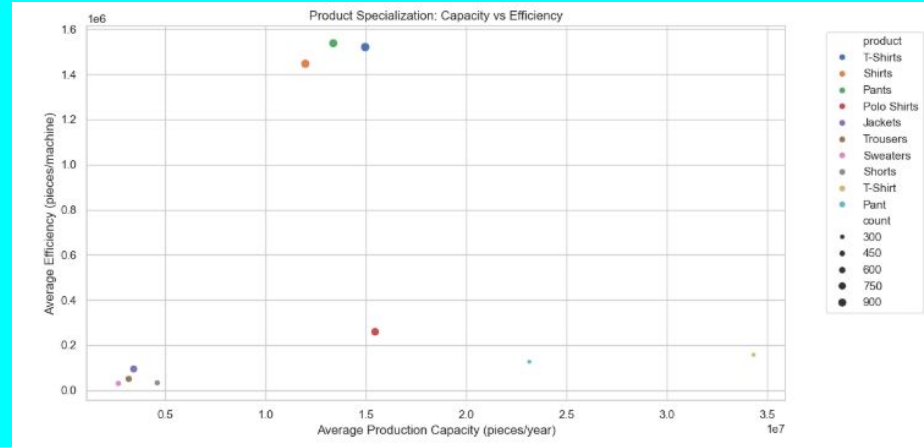
- Pants/Trousers: 1.5-1.8 pieces/machine 
- Shorts: 1.2 pieces/machine 

- 🐌 **Specialty Challenges:**

- Jackets: 0.8 pieces/machine (Complex) 
- Sweaters: 0.5 pieces/machine (Labor-Intensive) 

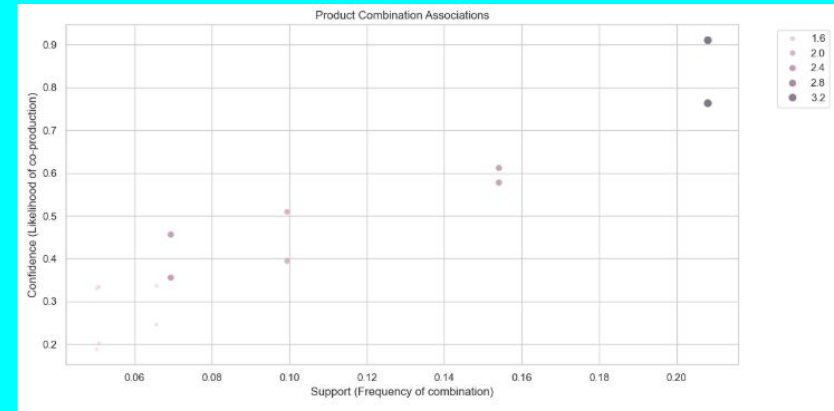
- 🎯 **Factory Goals:**

- T-Shirts: Aim for >3.0 pieces/machine  
- Woven: Target >2.0 pieces/machine  
- Sweaters: Automate!  



Product Pair-Ups: What Goes Together?

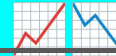
-  **Association Rules:**
 - Finds product combos that are often made together.
-  **Support (X-Axis):**
 - How often a combo happens.
 - High Support = Common combos.
-  **Confidence (Y-Axis):**
 - If you make A, how likely to make B?
 - High Confidence = Reliable co-production.
-  **Lift (Color/Size):**
 - Combo more likely than by chance?
 - $Lift > 1$ = Strong Link!
-  **Point Spread:**
 - Lower Left: Rare, unreliable combos.
 - Upper Right: Frequent, reliable combos.
-  **Support Range:**
 - 0.04 to 0.22 (Wide variation in combo frequency).
-  **Confidence Range:**
 - 0.2 to 0.9 (Huge difference in co-production likelihood).
-  **Lift Range:**
 - 1.6 to 3.2 (All combos are linked, but some are WAY stronger).
-  **Strong Associations:**
 - Upper Right = Key combos for planning.
-  **Weak Associations:**
 - Lower Left = Less important.



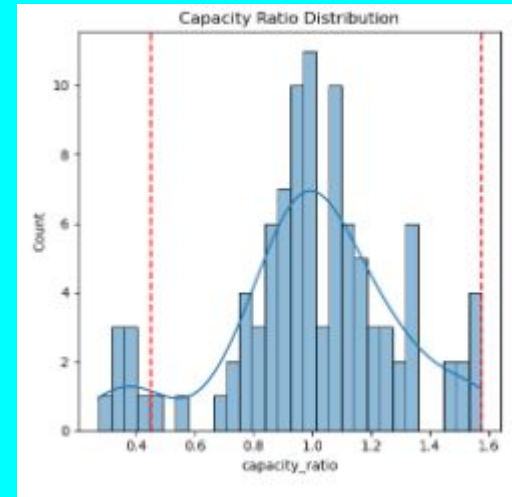
-  **Lift Power:**
 - Shows combos aren't random!
-  **Production:**
 - Focus on high-support/confidence.
 - Plan resources for co-production.
-  **Marketing:**
 - Cross-sell high-lift combos!
-  **Inventory:**
 - Predict demand based on combos.



Capacity Check: Who's Above Average?

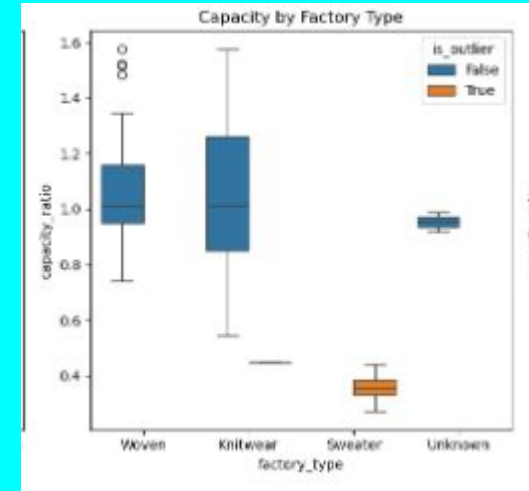


-  **Central Tendency:**
 - Roughly normal distribution around 1.0 ratio
 - Most factories (peak): 0.8-1.2 range
-  **Outlier Limits:**
 - Low: ~0.6 (Red line)
 - High: ~1.4 (Red line)
 - (Likely $1.5 * IQR$)
-  **Skew:**
 - Slight right skew (Longer high-ratio tail)
-  **"Normal" Range:**
 - 68% of factories: 0.8-1.2
- **** extremos:****
 - 8-10 factories at each end
-  **High Performers (Right):**
 - 6 factories > 1.4 ratio (Best practices?) 
 - Efficient operations
-  **Underperformers (Left):**
 - 8 factories < 0.6 ratio (Fix needed!) 
 - Upgrade equipment? Train workers?



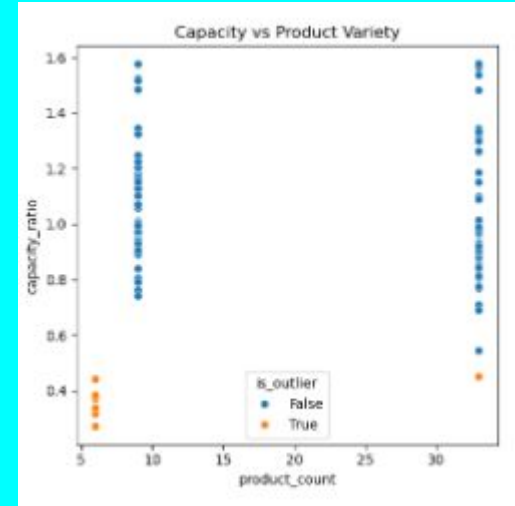
Type vs. Capacity: Who's the Most Efficient?

-  **Woven Factories:**
 - Median Capacity Ratio: ~1.0 (Benchmark) 🎯
 - Capacity Range: Wide (0.8-1.3) ↔
 - Most High & Low Performers ⭐📊
-  **Knitwear Factories:**
 - Median Capacity Ratio: ~1.1 (Higher) 📈
 - Capacity Range: Tight (0.9-1.2) ✅
 - Consistent Performance
-  **Sweater Factories:**
 - Median Capacity Ratio: ~0.7 (Lowest) 🐢
 - All Below Benchmark ⚠️
 - Potential Issues
- ⭐ **Outliers:**
 - Woven: Some Stars, Some Strugglers ⭐📊
 - Knitwear: Few Stars ⭐
 - Sweater: No Stars, All Struggling 😞
-  **For Buyers:**
 - Bulk: High-Capacity Woven 📦
 - Consistent Quality: Knitwear ✅
 - Specialty: Sweater (Longer Wait?) ⌚



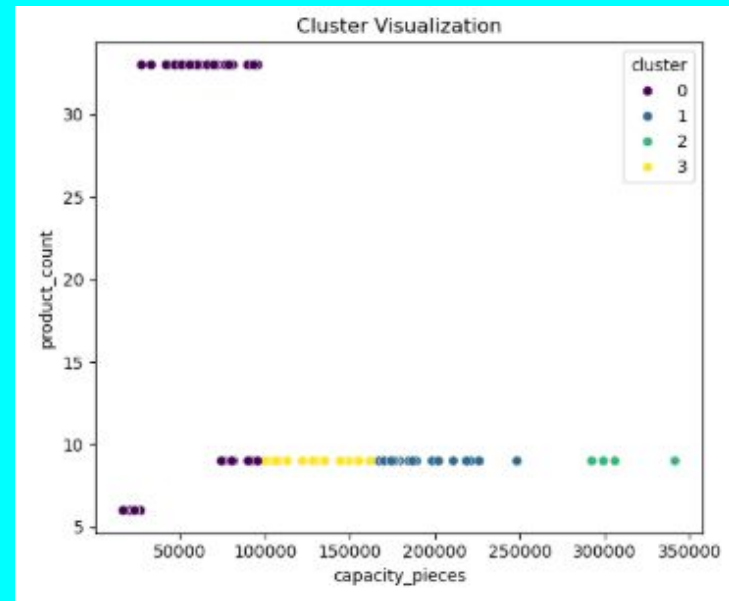
vs. : Is Variety Killing Capacity?

-  **General Trend:**
 - More products = Less capacity ratio
-  **High Capacity:**
 - <15 products: Best capacity (1.0-1.4 ratio)
-  **Sharp Decline:**
 - Capacity drops after 20 products
-  **High-Capacity Outliers:**
 - Only factories with <15 products have high capacity (>1.4 ratio)
-  **Low-Capacity Outliers:**
 - Across all product counts
 - Worst case: 5+ factories with <10 products AND low capacity (<0.6)
-  **Sweet Spot:**
 - 8-12 products: Peak performance (1.1-1.3 ratio)
-  **Drop-Off Point:**
 - After 15 products: 75% of factories <1.0 ratio
-  **High-Capacity Factories (>1.2 ratio):**
 - Keep 8-12 products
 - Add ≤2 new products/year
-  **Struggling Factories (<0.8 ratio):**
 - Cut product lines to ≤15 items
 - Focus on 3-5 high-margin products



Factory Clusters: Who Are They? What Do They Need?

-  **Clear Separation:**
 - 4 distinct factory groups! 
-  **Cluster 0 (Blue):**
 - High product count, medium capacity 
-  **Cluster 1 (Green):**
 - Low product count, high capacity 
-  **Cluster 2 (Red):**
 - Medium product count, low capacity 
-  **Cluster 3 (Orange):**
 - High product count, high capacity (Elite!) 
-  **Cluster Stats:**
 - Cluster 0: 15-25 products, 50K-150K pieces (Diversified Mid)
 - Cluster 1: 5-15 products, 200K-350K pieces (Focused High)
 - Cluster 2: 10-20 products, 10K-50K pieces (Generalist Low)
 - Cluster 3: 20-30 products, 200K-300K pieces (Diversified High)
-  **Cluster 2 Fix:**
 - Cut 5-7 products 
 - Boost capacity to $\geq 100K$ pieces 
-  **Cluster 1 Plan:**
 - Keep focused strategy 
 - Maybe add 2-3 products? 
-  **Cluster 0 Upgrade:**
 - Better scheduling 
 - New equipment 



Cluster Capacity: Who's the Biggest?

Capacity Hierarchy:

- Cluster 1 > Cluster 3 > Cluster 0 > Cluster 2 (Biggest to Smallest)

Median Capacity:

- Cluster 1: ~275,000 pieces (Top Dog!) 
- Cluster 3: ~200,000 pieces
- Cluster 0: ~125,000 pieces
- Cluster 2: ~75,000 pieces (Needs Help!) 

Consistency:

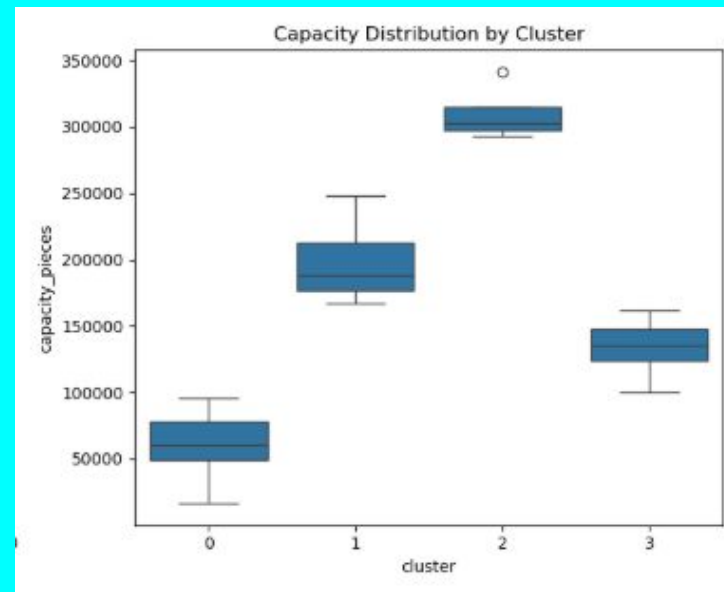
- Cluster 1: Tight IQR (250K-300K) = Steady High 

Variability:

- Cluster 2: Wide IQR (50K-150K) = Inconsistent 

Outliers:

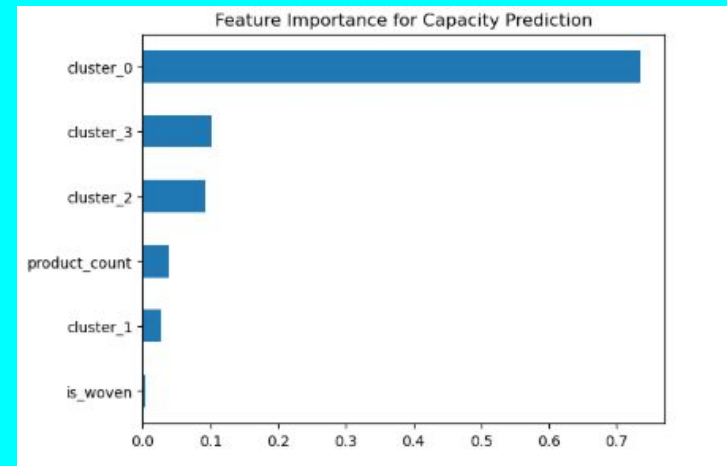
- Cluster 3: Some capacity giants (>350K) 





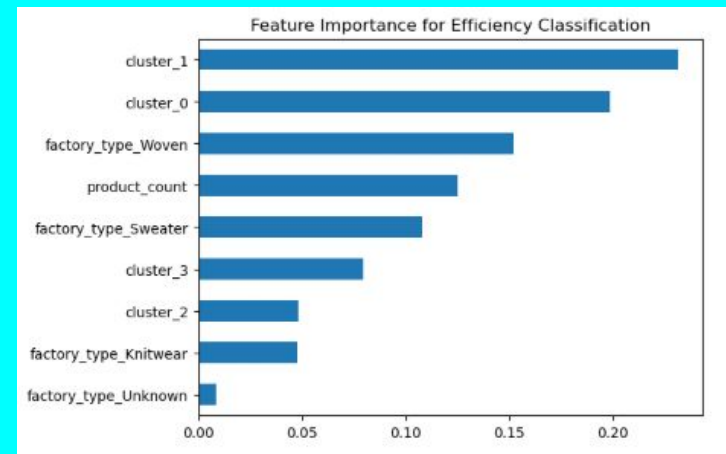
Capacity Predictors: What's Driving Output? 🏭

- 📦 **Top Dog: product_count (0.52):**
 - Most important! 52% of predictions!
 - More products = More output? 🧠
- 🧶 **Woven Power: is_woven (0.31):**
 - Woven factories are different!
 - 31% influence on predictions.
- 📍 **Cluster Clout (0.08-0.09):**
 - Clusters matter (location/ops).
 - Moderate impact.
- 👤 **Where Are the Others?**
 - Age, employee count? Not important!
- 🎯 **Big Takeaway:**
 - Variety (products) is KEY!
 - Woven factories are a major factor.
 - Clusters play a role.



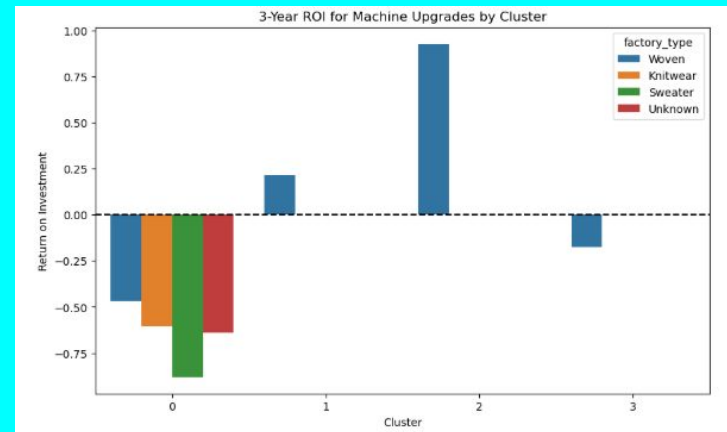
⚡ Efficiency Drivers: What Makes Factories Efficient? ⚙️

- 📌 **Cluster Kings:**
 - `cluster_1` (0.20) and `cluster_0` (0.15): Location/Operations are KEY! 👑
- 🧵 **Woven's Way:**
 - `factory_type_Woven` (0.12): Woven factories have a special efficiency style.
- 📦 **Product Power:**
 - `product_count` (0.08): Variety matters a bit.
- 📌 **Cluster Clout (Less):**
 - `cluster_3` (0.07) and `cluster_2` (0.05): These clusters are less unique in efficiency.
- 🧑 **Type? Nah:**
 - Sweater, Knitwear, Unknown: Don't really affect efficiency.
- 🎯 **Big Picture:**
 - Where you are and how you run things (clusters) are the BIGGEST factors.
 - Woven factories are different.
 - Product variety has a small effect.



💰 Upgrade ROI: Where Should We Invest? ⚙️

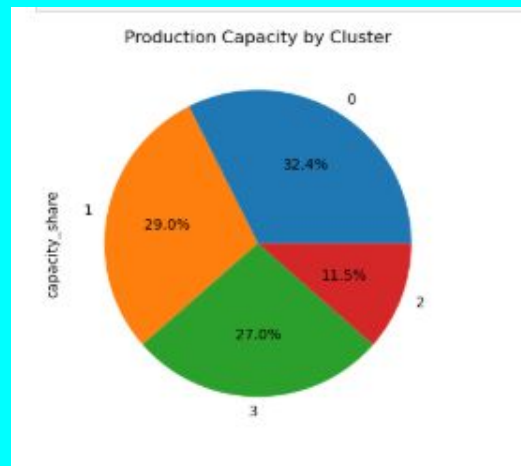
- 🔴 **Cluster 0:**
 - Strong ROI (0.75-1.00) for Woven & Knitwear! 🤖
- 🔴 **Cluster 1:**
 - Woven: Positive ROI (0.25) ✅
 - Knitwear: Negative ROI (-0.25) ❌
- 🔴 **Cluster 2:**
 - Negative ROI for ALL! 📉
 - Avoid upgrades here.
- 🔴 **Cluster 3:**
 - Woven: Marginal positive ROI (0.25) 🤔
- 🧵 **Woven:**
 - Best overall! ROI: -0.25 to 1.00
- 🧶 **Knitwear:**
 - ROI: -0.75 to 0.75 (Very risky!) 😬
- 🧶 **Sweater:**
 - Poor returns. ROI: -0.50 to 0.25
- ❓ **Unknown:**
 - Negative ROI in all clusters.
- 🚀 **Immediate Action:**
 - Cluster 0: Woven (1.00 ROI) & Knitwear (0.75 ROI)
- ⚠️ **Caution:**
 - Cluster 2: NO upgrades!
 - Cluster 1 Knitwear: Re-evaluate!



- 💰 **Capital:**
 - 70-80% to Cluster 0 Woven/Knitwear
 - \$0 to Cluster 2
- ⚙️ **Ops:**
 - Cluster 1 Knitwear: Fix root causes!
 - Cluster 0: Learn from the best!
- ⚖️ **Risk:**
 - Don't put all eggs in one basket!
 - Test upgrades first.

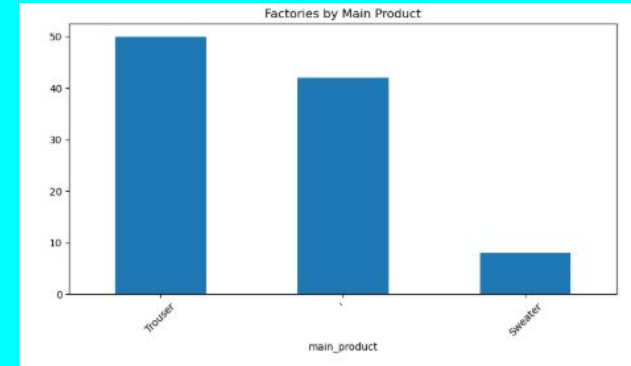
Cluster Capacity: Who's Making the Most?

-  **Cluster 0:**
 - Largest: 32.4% of total capacity!
-  **Cluster 3:**
 - Close second: 29.0%
-  **Cluster 2:**
 - Smallest: Only 11.5%
-  **Capacity Imbalance:**
 - Top 2 (0 & 3) = 61.4% of total!
 - Cluster 2: Underutilized! 😞
-  **Risk Alert:**
 - Over 60% in 2 clusters = Vulnerable! 💣



Product Focus: Too Many Eggs in One Basket?

-  **Top Product Risk:**
 - One product: ~50 factories (Over-reliance!)
-  **Second Product Risk:**
 - Second product: ~40 factories (Still concentrated)
-  **Long Tail:**
 - Beyond top 2: Steep drop-off (≤ 30 factories)
 - Niche products: Vulnerable?
 - Underserved products: Opportunity! 
-  **Missing Context:**
 - Product names HIDDEN! (Only "main_product" & "generator")
 - Can't tell if it fits Bangladesh's strengths (knit vs. woven)
 - Can't pinpoint high-risk items!
-  **Action Needed:**
 - Get those product names!



Factory Check-Up: Key Takeaways!

-  **Cluster Focus:**
 - Gazipur & Dhaka dominate! (61.4% capacity) 🏰
 - Cluster 2 is underutilized! (11.5%) 😞
-  **Woven & Knit Rule:**
 - 83.4% of factories! 💪
 - Woven leads in capacity. 🧵
-  **Capacity Range:**
 - Huge variation: 10^5 to 60M pieces/year! 🧐
-  **Young Industry:**
 - Most factories post-1980s. 🧒
-  **Machines Matter:**
 - More machines = More output! 🤖
-  **Product Concentration:**
 - Top 2 products = ~45% of factories. ⚠️
- ⚡ **Efficiency Varies:**
 - Woven: High potential (and risk) ⚡
 - Sweater: Lags. 🐢
- 💰 **Upgrade ROI:**
 - Cluster 0 Woven/Knit = Best ROI! 🚀



THANK YOU!